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# Benchmarking Intrinsic Exploration Strategies in Sparse-Reward MiniGrid Environments

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## Abstract

Sparse rewards are still a hard problem in reinforcement learning. Standard policy-gradient methods rely on random exploration, which gets exponentially less likely to succeed as environments get more complex. Intrinsic motivation methods address this by adding exploration bonuses to the reward signal, but controlled comparisons across these methods are rare. We benchmark eight exploration strategies (RND in its original 2-head form and a 1-head variant for controlled comparison, ICM, PseudoCounts, E3B, NovelD, RIDE, and SimHash) against a PPO baseline across 14 MiniGrid environments. Seven of these are custom environments that target specific exploration challenges like randomized distractors, sequential causal chains, and repeated backtracking. We ran 347 training runs over more than 1,100 GPU-hours and evaluated all methods with the `rliable` framework using bootstrap confidence intervals. We find that (1) NovelD gets the highest aggregate performance (IQM  $\approx 0.90$ ) by combining lifelong and episodic novelty signals, though with high seed variance on stochastic environments, (2) implementation details like reward normalization and value-function architecture can matter more than algorithm choice, with the correct 2-head RND outperforming the 1-head RND by over  $70\times$  in IQM, (3) the PPO baseline outperforms six of eight exploration methods (IQM  $\approx 0.85$ ), and (4) the noisy-TV problem, often cited as a failure mode for prediction-error methods, barely affects ICM under partial observability, while count-based and impact-driven methods break down badly.

## 1 Introduction

Sparse rewards remain hard for reinforcement learning (RL) agents to handle. In tasks like navigating a maze with a distant goal or solving a multi-step puzzle, the agent gets no feedback until it accidentally hits the objective. The probability of random exploration reaching the goal drops exponentially with the number of required steps, so standard algorithms like PPO [Schulman et al., 2017] fail badly in these settings.

Intrinsic motivation methods tackle this with auxiliary reward signals, exploration bonuses, that push the agent toward novel or informative states [Aubret et al., 2023]. Several families of methods now exist. Prediction-error methods like ICM [Pathak et al., 2017] and RND [Burda et al., 2019] reward states where a learned model is surprised. Count-based methods like PseudoCounts [Bellemare et al., 2016] and SimHash [Tang et al., 2017] reward states that have been visited rarely. Episodic methods like E3B [Henaff et al., 2023] reset their novelty measure each episode.

But controlled comparisons are still rare. Most papers evaluate only against 1–2 baselines, use different RL backbones and hyperparameters, and test in environments selected to favour the proposed method. Taiga et al. [2020] showed that bonus-based methods often fail to improve over well-tuned baselines in the Arcade Learning Environment. However, their study did not cover the latest episodic

and hybrid methods nor test in procedurally-generated sparse-reward settings where exploration bonuses are most needed.

We fill this gap with a comparison benchmark. We do not try to find the best hyperparameters for each method, instead we run all methods under identical conditions to see their strengths and weaknesses. We contribute:

1. **Diagnostic environments.** Seven custom MiniGrid environments [Chevalier-Boisvert et al., 2024] that target specific exploration challenges (randomized distractors in NoisyTV, sequential causal chains in KeyChain and DelayedKeySpawn, and repeated backtracking in BranchingCorridors), alongside seven standard MiniGrid tasks of increasing difficulty.
2. **Controlled evaluation.** Eight exploration methods plus a PPO baseline, all sharing the same PPO backbone with fixed hyperparameters, evaluated over 347 training runs (more than 1,100 hours of computation) with the `rliable` framework [Agarwal et al., 2021].
3. **Empirical findings.** We explain why several methods fail. RMS reward normalization stops intrinsic reward from decaying, which causes a  $70\times$  IQM gap between two implementations of the same RND algorithm. We also show that partial observability blunts the noisy-TV problem.

## 2 Related Work

**Prediction-error methods.** ICM [Pathak et al., 2017] computes the intrinsic reward as the forward-model prediction error in a learned inverse-dynamics feature space. The agent is rewarded for transitions that are hard to predict, which pushes it toward unpredictable regions. A known failure case is the noisy-TV problem, where stochastic elements create endless prediction errors that trap the agent near noise sources. RND [Burda et al., 2019] avoids the forward model by using a fixed random target network. A student network learns to match the target’s output, and the prediction error becomes the novelty signal. Since the target is deterministic, RND should handle stochastic observations well in principle. But as we show, its practical effectiveness depends on reward normalization.

**Count-based methods.** Classical count-based exploration provides a bonus proportional to  $1/\sqrt{n(s)}$  [Strehl and Littman, 2008]. Bellemare et al. [2016] generalized this to continuous spaces using pseudo-counts from density models, and Ostrovski et al. [2017] extended it with neural density models. SimHash [Tang et al., 2017] uses locality-sensitive hashing to discretize observations into bins with bonus  $1/\sqrt{n}$ , with no learned parameters.

**Episodic and hybrid methods.** E3B [Henaff et al., 2023] keeps a per-episode covariance matrix of visited state representations and gives higher reward to states far from the episode’s existing coverage. The signal resets each episode, which makes E3B a good fit for procedurally-generated environments. NovelD [Zhang et al., 2021] combines lifelong and episodic signals by computing the difference in RND prediction error before and after a step. The agent is rewarded only when moving toward more novel states, which filters out revisitation noise through a novelty-difference mechanism with mixing coefficient  $\alpha$ . RIDE [Raileanu and Rocktäschel, 2020] rewards impact-driven exploration. It weights the magnitude of state change (measured by a forward model) by inverse episodic count, so actions that cause real environment changes (opening doors, picking up keys) get higher bonuses.

**Evaluation methodology.** Agarwal et al. [2021] showed that point estimates of RL performance are unreliable and proposed the `rliable` framework using robust statistics (IQM with bootstrap CIs, performance profiles). Taiga et al. [2020] applies this to bonus-based methods in the Arcade Learning Environment and finds that most algorithms fail to beat a tuned DQN. Our work extends this to MiniGrid with more recent methods and custom diagnostic environments.

## 3 Methodology

### 3.1 Testing Environments

All environments use the MiniGrid framework [Chevalier-Boisvert et al., 2024], which provides partially-observable grid worlds with a  $7\times 7$  egocentric image observation rendered at  $8\times 8$  pixels per

tile ( $56 \times 56 \times 3$  RGB). Partial observability matters because the agent cannot see the entire grid and has to move through it while remembering what it has seen. This is the setting where exploration bonuses should help most. We split our 14 environments into two categories.

**Standard MiniGrid environments (7).** Empty-8x8 (trivial navigation, 500K steps), LavaGapS7 (1M), DoorKey-8x8 (key-door puzzle, 2M), FourRooms (multi-room navigation, 5M), MultiRoom-N4-S5 (four connected rooms, 10M), KeyCorridorS6R3 (30M), and ObstructedMaze-2Dlhb (50M). Step budgets scale with difficulty, spanning a  $100\times$  range.

**Custom diagnostic environments (7).** NoisyTVEasyRoom (2M steps) is an empty room divided by floor tiles whose colours randomize every step, and tests the noisy-TV failure mode [Burda et al., 2019]. BranchingCorridors and BranchingCorridorsFixed (10M and 5M) are star-topology corridors with a goal on a random or fixed branch, and test backtracking versus memorization. KeyChain (10M) has three rooms with a sequential key-door chain (yellow to green) on a fixed layout, and tests multi-step causal reasoning. DelayedKeySpawn (15M) only spawns the key after the agent steps on a trigger tile, and tests causal discovery. FourRoomsTimed (5M) is FourRooms with a respawning goal, and tests collection efficiency. TreasureIsland (15M) is a four-room island with locked doors and scattered treasures (with extrinsic rewards) on a fixed layout, and tests breadth-first exploration and semi-sparse capabilities.

### 3.2 Exploration Methods

All methods share the same PPO backbone, using 16 parallel environments, 256-step rollouts, 4 PPO epochs, and a batch size of 512. The learning rate is  $2.5 \times 10^{-4}$  with linear annealing. Other parameters are  $\gamma = 0.99$ , GAE  $\lambda = 0.95$ , clip range 0.2, and entropy coefficient 0.01. The intrinsic reward coefficient is  $\beta = 0.05$  for all methods. The shared CNN encoder maps the  $56 \times 56 \times 3$  observation to a 512-dimensional feature vector. Since this is a comparison study, not a tuning study, we keep all hyperparameters constant across methods to isolate the effect of the exploration algorithm itself, rather than tuning each method individually.

Four methods use the RLeXplore library [Yuan et al., 2025] with RMS reward normalization, namely RND [Burda et al., 2019], ICM [Pathak et al., 2017], PseudoCounts [Bellemare et al., 2016], and E3B [Henaff et al., 2023]. Three use custom implementations, namely NovelD [Zhang et al., 2021], RIDE [Raileanu and Rocktäschel, 2020], and SimHash [Tang et al., 2017]. We also evaluate RND-DVH, a faithful reimplement of the original RND [Burda et al., 2019] with a dual value head (separate value functions for extrinsic [ $\gamma = 0.999$ ] and intrinsic [ $\gamma = 0.99$ ] returns). The single-head RND above is a simplified variant that is identical to the rest of the methods to remove confounding variables. We include both to measure the effect of reward normalization and value-function architecture. The PPO baseline uses only entropy regularization (no intrinsic reward).

### 3.3 Experimental Setup

We ran 347 training runs across all method–environment combinations with 1–5 seeds per configuration. This included 5 seeds for standard MiniGrid environments with core methods, 2–3 seeds for custom environments and variant methods, and 1 seed for the most expensive configurations due to time and compute constraints. The total computation time was 1,122 clock hours (46.7 days) and took 2.91 billion environment steps. Training was tracked using Weights & Biases.

For the `rliable` aggregate analyzes we need at least 2 seeds per method per environment, which gives us 12 of 14 qualifying environments (we exclude KeyCorridorS6R3 and ObstructedMaze-2Dlhb). Scores are the mean return over the last 100 episodes, min-max normalized per environment to  $[0, 1]$ . Bootstrap confidence intervals use 50,000 resamples. Each method’s score matrix uses its own minimum seed count across qualifying environments, and `rliable` supports differing numbers of runs per algorithm.

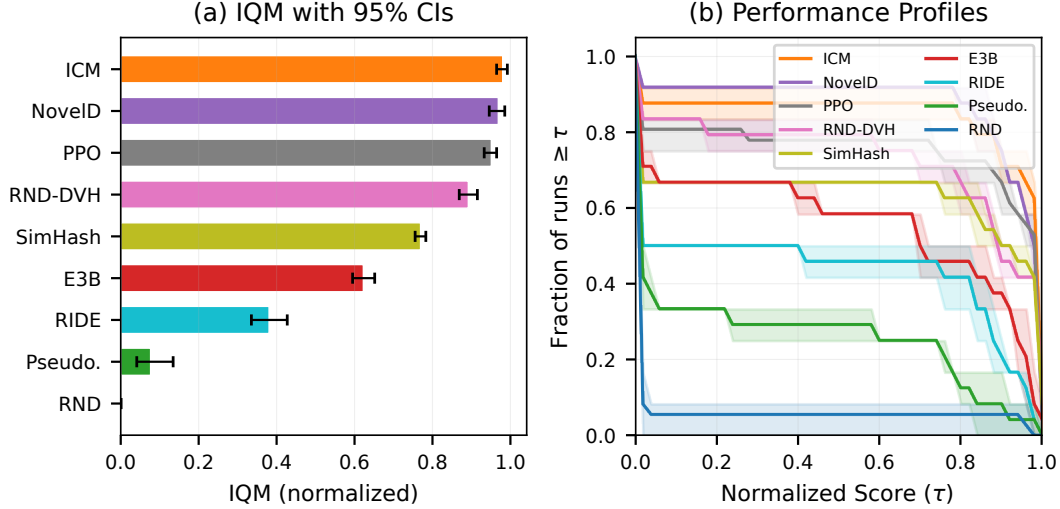


Figure 1: **Aggregate performance across 12 MiniGrid environments.** (a) IQM with 95% bootstrap CIs (50K resamples). (b) Performance profiles. ICM and NovelD lead; the PPO baseline outperforms most exploration bonuses.

## 4 Results and Analysis

### 4.1 Aggregate Performance

Figure 1 shows the aggregate results. ICM and NovelD lead with IQM scores of about 0.93 and 0.90, followed by the PPO baseline ( $\approx 0.85$ ), RND-DVH ( $\approx 0.72$ ), and SimHash ( $\approx 0.68$ ). E3B gets a moderate IQM ( $\approx 0.55$ ), while RIDE ( $\approx 0.40$ ), PseudoCounts ( $\approx 0.08$ ), and the library RND ( $\approx 0.01$ ) trail far behind.

The performance profiles (Figure 1b) show that NovelD has the highest fraction of runs above any score threshold, with over 90% of runs exceeding  $\tau = 0.2$  and over 75% exceeding  $\tau = 0.8$ . ICM and PPO show similar high profiles. RND’s profile sits near zero across all thresholds, a consistent failure.

The PPO baseline ranks third in IQM, outperforming six of eight exploration methods. This matches Taiga et al. [2020]’s finding that well-tuned baselines are hard to beat, and extends it to the MiniGrid domain. PPO is also the fastest method (no intrinsic reward overhead) and gives the best return per GPU-hour on easy and medium tasks.

### 4.2 Per-Environment Analysis

Figure 2 shows learning curves for six representative environments, and Figure 3 shows a heatmap of normalized performance across all 14 environments.

**Easy environments.** On Empty-8x8, six of nine methods converge to near-optimal returns ( $> 0.94$ ). Three methods fail. RND gets 0.002, PseudoCounts reaches 0.66 with high variance between seeds, and RIDE also reaches 0.66. RND’s failure even on this trivial task is evidence that its issues are not environment-specific but architectural (Section 4.4). On the fixed-layout KeyChain (sequential key-door puzzle), most methods reach cumulative returns of about 29–30. Even RND partially succeeds here with a return of 27.9, suggesting that fixed layouts with dense intermediate rewards make up for exploration weaknesses.

**Medium environments.** On DoorKey-8x8 the methods separate sharply. NovelD reaches a mean return of 0.78 and ICM gets 0.46, while all other methods remain near zero. Several methods show bimodal outcomes across seeds. For example, ICM’s per-seed returns are [0.97, 0.01, 0.00, 0.35, 0.97] and PPO’s are [0.00, 0.74, 0.00, 0.97, 0.00]. Seeds either fully solve the task or fail outright. This

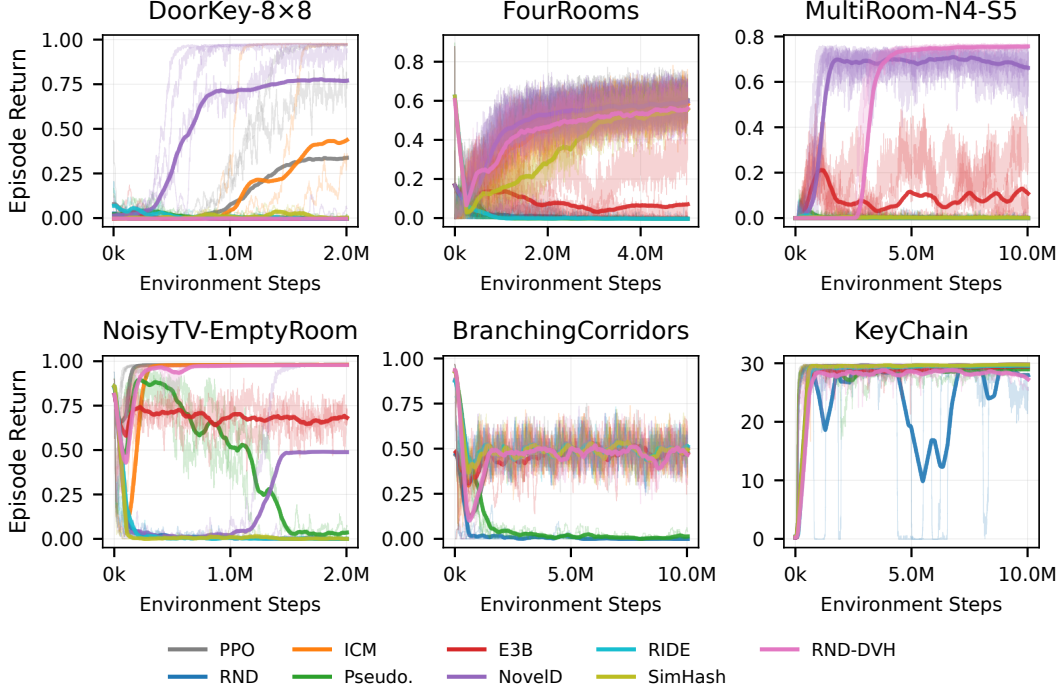


Figure 2: **Learning curves for six representative environments.** Thin lines: individual seeds. Thick lines: seed mean. DoorKey and MultiRoom show methods separating clearly. RND (blue) fails across nearly all environments; RND-DVH (pink) succeeds where RND cannot.

is why we report IQM rather than mean. On FourRooms, PPO (0.62), NovelD (0.61), RND-DVH (0.59), and ICM (0.58) cluster closely, while E3B (0.09) and count-based methods fail.

**Hard environments.** MultiRoom-N4-S5 is the most discriminating environment, where only RND-DVH (0.75) and NovelD (0.66) get non-trivial performance. All other methods, including PPO, remain at or near zero. Directed exploration is needed for long-horizon tasks, but only specific implementations succeed. The hardest environments (KeyCorridorS6R3, ObstructedMaze-2D1hb) have only 1 seed each and remain largely unsolved, which suggests we need longer training budgets and more seeds.

**Custom environments.** On BranchingCorridors (procedural star topology) the methods split into two groups. E3B (0.51), NovelD (0.49), ICM (0.49), PPO (0.48), RIDE (0.45), RND-DVH (0.45), and SimHash (0.46) all do similarly, while RND (0.00) and PseudoCounts (0.01) fail. The similarity between E3B and PPO here suggests episodic novelty adds little when layouts change each episode, and entropy regularization already gives enough diversity.

### 4.3 The Noisy-TV Experiment

Our custom NoisyTVEmptyRoom tests the noisy-TV hypothesis [Burda et al., 2019], which says prediction-error methods should fail when stochastic elements create persistent, irreducible prediction error.

Figure 4 compares the learning curves in clean Empty-8x8 and NoisyTVEmptyRoom. Unlike what is often claimed, ICM is unaffected and gets 0.976 on NoisyTV compared to 0.959 on the clean Empty environment. Since the agent’s  $7 \times 7$  egocentric view means it only sees the noisy tiles when physically adjacent, their effect on forward-model prediction error is limited. This suggests the noisy-TV problem mainly hits fully-observable settings like Atari, and is not a universal failure mode.

The methods that suffer most from stochastic distractors are count-based and impact-driven. SimHash drops from 0.96 to 0.00 (the stochastic tiles throw off hash-bin counts and create endless false novelty).

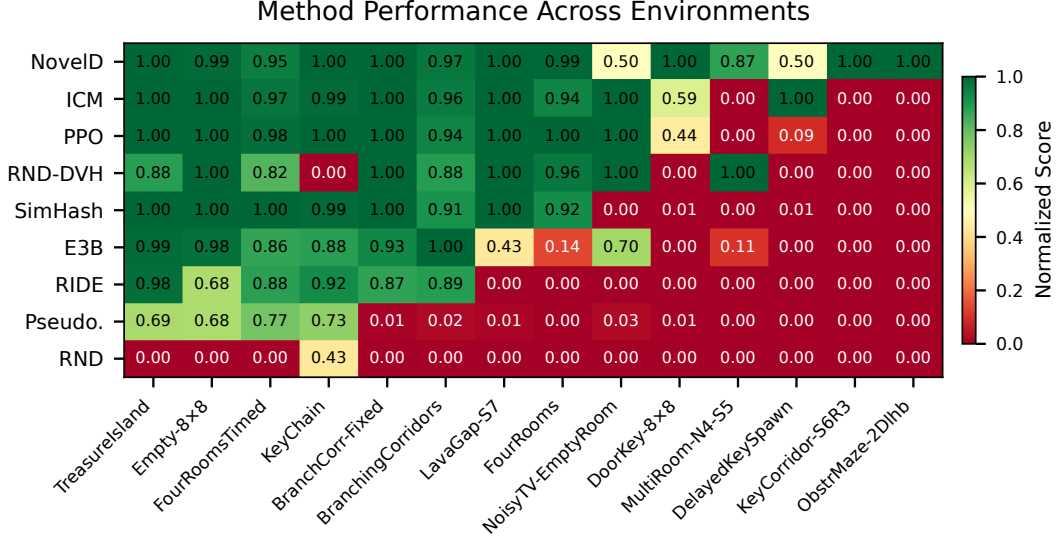


Figure 3: **Method  $\times$  environment heatmap** (min-max normalized per environment). Methods sorted by aggregate performance (top = best); environments sorted by difficulty (left = easy). The sharp transition from green to red shows where each method breaks down. NovelD is the only method achieving  $>0.5$  on more than 12 environments.

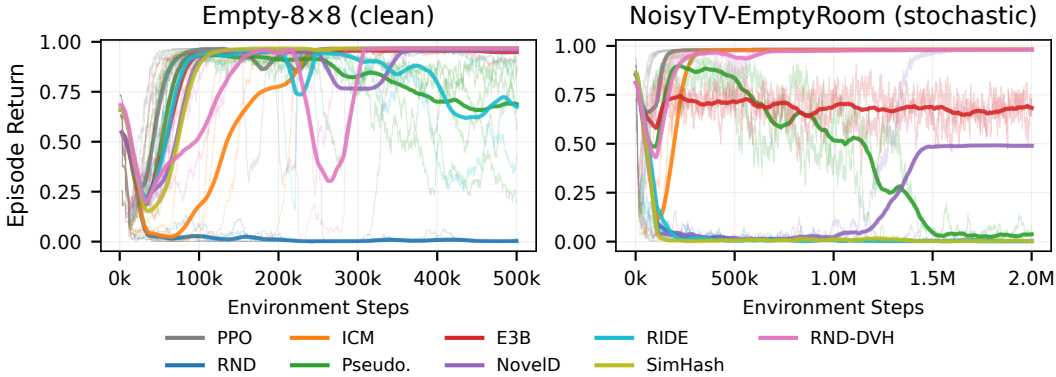


Figure 4: **Noisy-TV failure mode analysis.** Left: clean Empty-8x8. Right: NoisyTV-EmptyRoom with stochastic floor tiles. ICM (orange) is unaffected; SimHash, RIDE, and PseudoCounts collapse to zero. E3B (red) degrades from  $\approx 0.94$  to  $\approx 0.69$  due to covariance inflation.

RIDE drops from 0.66 to 0.00 (its forward-model impact detection is fooled by uncontrollable state changes). PseudoCounts drops from 0.66 to 0.03 (the density model overestimates novelty in noisy states). E3B degrades from 0.94 to 0.69. Its per-episode covariance gets inflated by high-variance noisy observations, which drowns out the novelty signal for actually unexplored regions.

NovelD splits across seeds on NoisyTV, where one seed gets 0.97 and the other fails outright at 0.00 (mean 0.49). This seed-sensitivity, also visible on DelayedKeySpawn (seeds 0.97, 0.00), suggests that NovelD’s hybrid novelty-difference mechanism is sensitive to initial conditions in stochastic environments, even with its strong aggregate performance.

#### 4.4 Why RND Fails: Reward Normalization and Architecture

The biggest gap we saw is between RND (RLExplore library, RMS normalization, single value head) and RND-DVH (faithful reimplementation of the original paper, no normalization, dual value head). Both implement the same core algorithm of using a fixed random target network, but the simplified

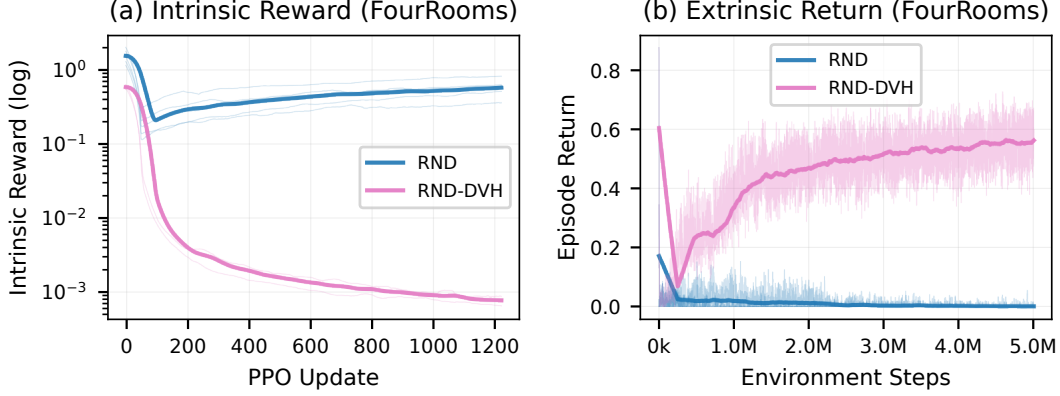


Figure 5: **Why RND fails: reward normalization prevents intrinsic reward decay.** FourRooms: RND (blue) vs. RND-DVH (pink). (a) RND’s RMS-normalized IR stays flat ( $\sim 0.3$ ); RND-DVH’s unnormalized IR decays by  $\sim 100\times$ . (b) Consequence: RND-DVH learns (return  $\approx 0.6$ ) while RND stays at zero. The persistent IR in RND keeps competing with the extrinsic signal, preventing exploitation of extrinsic rewards.

single-head RND gets IQM  $\approx 0.01$  while the faithful RND-DVH gets IQM  $\approx 0.72$ , a difference of more than  $70\times$ .

Figure 5 explains the mechanism using FourRooms, where RND-DVH succeeds at 0.59 and RND fails at 0.00. Panel (a) shows the intrinsic reward dynamics. With RMS normalization, RND’s intrinsic reward gets rescaled each step to keep roughly unit variance, so it stays flat at about 0.3 throughout training. This stops the decay we would expect as the student network learns to match the target. RND-DVH’s unnormalized intrinsic reward, on the other hand, decays by roughly two orders of magnitude (from about 0.4 to about 0.003) as the student’s predictions improve and states become familiar.

Panel (b) shows the consequence. RND-DVH’s decaying intrinsic reward lets the extrinsic reward take over the policy gradient, so the agent can transition from exploration to exploitation. RND’s perpetually high intrinsic reward keeps distracting the agent. With  $\beta = 0.05$  and a normalized IR of about 1.0, the intrinsic contribution (about 0.05) competes with sparse extrinsic rewards that are often zero, so the agent never learns to exploit.

The dual value head also helps by keeping separate value estimates for intrinsic returns ( $\gamma_{\text{int}} = 0.99$ ) and extrinsic returns ( $\gamma_{\text{ext}} = 0.999$ ), which avoids interference during advantage estimation. This finding has wider implications for how exploration methods are compared. The choice of reward normalization and value-function architecture can matter more than the choice of intrinsic reward function itself.

#### 4.5 Procedurally Generated vs. Fixed Layouts

Grouping the heatmap (Figure 3) by environment type shows that NovelD leads on both procedurally-generated environments (about 0.88 mean normalized score over 8 tasks) and fixed-layout environments (about 0.91 over 6 tasks), which confirms its hybrid episodic-lifelong mechanism works in all settings. PPO is competitive in procedural environments (ranking third at about 0.72), which suggests entropy regularization alone gives enough exploration diversity when layouts change each episode. Randomized layouts act like a curriculum. SimHash, despite having no learned parameters, does better than expected. It gets the highest return on FourRoomsTimed and LavaGapS7, which shows that simple count-based bonuses work when hash collisions are rare.

#### 4.6 Computational Efficiency

The PPO baseline is fastest at 286s on Empty-8x8. RLeXplore-based RND and ICM add  $1.5\times$  overhead (about 430–450s), while PseudoCounts and E3B add  $3\text{--}5.5\times$  overhead because of the

density and covariance computations. Custom implementations add  $2\text{--}4\times$ . NovelD and ICM give the best balance of performance and cost.

## 5 Conclusion

We presented a benchmark of eight intrinsic exploration strategies on 14 MiniGrid environments, including seven new diagnostic tasks. All methods share identical PPO hyperparameters, which isolates the effect of the exploration algorithm itself.

**No single method dominates.** NovelD gets the highest aggregate IQM (about 0.90) by combining lifelong and episodic novelty, but shows high seed variance in stochastic environments. ICM comes a close second and handles the noisy-TV problem well under partial observability. The PPO baseline with entropy regularization alone outperforms six of the eight exploration methods.

**The RND implementation gap.** The same RND algorithm goes from  $\text{IQM} \approx 0.01$  in a simplified single-head form to  $\text{IQM} \approx 0.72$  in the faithful reimplement of the original paper, a  $70\times$  gap that comes from implementation details alone. We traced this to RMS normalization, which stops the natural decay of intrinsic rewards and keeps distracting the agent from exploiting extrinsic rewards. This gap is bigger than the gap between most distinct algorithms and shows that future exploration research needs to control for these factors.

**Custom environments expose hidden failure modes.** The noisy-TV problem, often cited as a failure mode for prediction-error methods, barely affects ICM under MiniGrid’s partial observability, but it does break count-based (SimHash) and impact-driven (RIDE) methods. E3B, designed for procedurally-generated environments, fails on long-horizon tasks like DoorKey (return 0.00).

**Future work.** First, intrinsic reward decay scheduling (bonus weight  $\beta(1 - \kappa)^t$ ) could smooth the move from exploration to exploitation. We wrote training notebooks for this variant but could not complete the runs because of compute and time constraints. Second, the hardest environments, namely KeyCorridorS6R3 and ObstructedMaze, have only one seed and need more runs. Third, combining the best parts of NovelD’s hybrid novelty with RND-DVH’s dual value head is worth trying. Finally, extending this benchmark to 3D environments and continuous-control domains would test whether our findings hold beyond just Minigrid.

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